MILITARY SPECIFICATION

FIXTURE, LATHE, METAL CUTTING

This specification is approved for use by the Armament Research and Development Engineering Center (Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a holding fixture to support and retain work-pieces in a 13-inch lathe while milling, drilling, slotting, tapping, and boring operations are performed by the lathe. The fixture provides an indexing capability and a vertical feed capability for layout work, when used in conjunction with the lathe cross slide.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATINS

FEDERAL

PPP-P-40 - Preservation and Packing of Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

STANDARDS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FED-STD-H28 - Screw-Thread Standards For Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1189 - Standard Department of Defense Bar Code Symbology
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The fixture shall be new and of the current standard commercial design meeting the performance requirements of this specification. All parts subject to wear or replacement shall be accessible for adjustment or repair. The fixture shall provide a means of holding round and rectangular workpieces, and shall include a graduated vertical feed. The fixture shall perform as specified herein when mounted on a 13-inch lathe.

3.2.1 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.2.2 Threads. The design, manufacture, and configuration of threads shall conform to FED-STD-H28, and the applicable Detailed Standard section referenced therein.

3.2.3 Interchangeability. All parts of the fixture shall be manufactured to definite standards and tolerances. All parts of the fixture shall be interchangeable without requiring modification of the replaced part or its mating components. All parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer's part number and shall be identified in the

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repair parts manual. All replacement, repair, and spare parts shall provide the same or better form, fit, function, and performance as the parts provided initially installed in the fixture. When industry or Government standards are applicable, all replacement parts shall conform to the standards.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry. When dissimilar metals are in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The fixture shall be furnished complete so that, when installed, it can be used for all the operations for which it was designed. Only that reclamation of defective parts or assemblies which retains original strength, performance, interchangeability, appearance, and dimensions shall be allowed.

3.4.1 Castings and forgings. Castings shall be free of blowholes, porosity, hard spots, shrinkage defects, cracks, and other defects. Forgings shall be free of scale, enclosures, cold shuts, mismatching, and other defects that could affect structural strength. The physical and chemical properties of castings and forgings shall be adequate to meet the performance requirements of this specification.

3.4.2 Sizes and capacities. The fixture shall be a size that is adaptable for use on 13-inch lathes. The fixture shall have a vertical adjustment of not less than 3 1/2 inches. The working surface (fixture plate) shall not be less than 5 by 5 inches. The fixture shall be capable of holding round stock up to two inches in diameter, and square stock up to two inches.

3.5 Performance and accuracy characteristics.

3.5.1 Fixture. The fixture shall perform as specified herein, when mounted on a 13-inch lathe. Vertical adjustment shall be accomplished in increments of 0.001 inch. An error in adjustment shall not exceed 0.0005 of an inch per inch of adjustment. The working surface (fixture plate) shall be hardened (45 to 48 on the Rockwell C scale) to prevent wear and shall be finish ground.

3.5.1.1 Key seat cutting. The fixture shall hold round bar stock for key seat cutting. The surfaces of the cuts shall be free of chatter marks.

3.5.1.2 Keyway cutting. The fixture shall securely hold round bar stock while cutting keyways in one continuous operation. Keyway cuts shall not vary more than .003 inch per foot from a straight line.

3.5.1.3 Vertical slotting. The fixture shall be capable of holding the workpiece while milling a vertical slot. Milling shall be accomplished by using a vertical feed on the fixture and an end mill in the lathe. The accuracy of a predetermined

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slot width shall be plus/minus 0.001 inch. The slot shall be perpendicular to the horizontal plane and shall have an accuracy of plus/minus 0.0015 per inch of slot length.

3.5.2 Indexing device. The indexing device shall be graduated in degrees and shall accommodate round bar stocks from one-fourth inch in diameter to two inches in diameter. The device shall be capable of being secured in any position from 0 through 360 degrees with an accuracy of plus/minus 30 minutes when measured from the base cut.

3.5.3 Rigidity. The rigidity of the fixture shall produce a 125 micro-inch arithmetical average (AA) maximum finish on a one inch square bar of AISI 1045 cold rolled steel after a cut of one-half inch depth, full width, for a distance of 2 1/2 inches, at a rate of 1 3/4 inches per minute (minimum) using a one inch diameter end mill.

3.6 Protective finish. The protective finish shall be paint. Surfaces to be painted shall be cleaned, free of all foreign matter, and dried immediately prior to painting. The protective paint coating shall be at least 2.0 roils (0.002 inch) thick. The paint adhesion shall be such that no paint shall be capable of being peeled by fiberglass tape applied over two intersecting cuts of not less than 2 inches exposing the base metal. The surface hardness of protective finish shall be between the industrial standards of H to HB lead. Where no protective finish is utilized (such as bedways, machine guide surfaces, etc.) the natural finish of the material or the finish obtained from heat treatment is permissible provided the surfaces are free from scale or corrosion.

3.7 Standard equipment and accessories. The fixture shall be furnished with the necessary standard equipment (see 6.2.1) to permit mounting on lathes, including all the necessary work holding clamps and fasteners needed to perform the operations specified herein.

3.7.1 Case. A case shall be furnished to carry and store the fixture, including any required attachments, special wrenches, spare parts, instructions, or other accessory items. The case shall be provided with means to secure the individual items so that they cannot be displaced during normal handling. Brackets, clamps, contours, or other holding and restraining devices shall be provided as required. The case shall be fabricated from either sheet steel or high density polyethylene (see 3.7.1.1 and 3.7.1.2).

3.7.1.1 Steel case. The steel case shall be manufactured from no less than 21 gage (0.033 inch thick) steel sheet, with a hinged lid and a positive type fastening latch.

3.7.1.2 High density polyethylene case. The polyethylene case shall be constructed of high density polyethylene (specific gravity of 0.941 to 9.965), which is blo molded or injection molded to the desired shape, with a contoured interior to retain each component. The case shall have a carrying handle and positive closure clasps. The case shall have a lid that opens on a mechanical style hinge, consisting of a male plastic pin fitting into a female dent, forming a swivel socket configuration. Living hinges, hinges consisting of a flexible flap of polyethylene extending from lid to base, are not acceptable. No metal parts shall be used in the construction of the case, including the handle, clasps, hinge, feet, markings or fasteners.

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3.7.2 Special wrenches. All special wrenches or tools that are required for operation or maintenance of the fixture shall be furnished.

3.8 Product identification. Nameplate(s) permanently and legibly marked in accordance with MIL-STD-130, including serial number, shall be securely attached to each fixture and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall assure that the lathe fixture shall have the efficient operating characteristics found in standard commercial units and as specified in Section 3 herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of a. through c. below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.5).

b. Tests (see 4.6).

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c. Packaging inspection (see 4.7).

4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105 and the following:

INSPECTION/TEST	INSPECTION LEVEL	AQL
Product examination (4.5)	General Level II	1.0
Tests (4.6)	General Level II	1.0
Packaging inspect ion (4.7)	Special Level S-4	4.0

4.5 Product examination. Visually and manually examine the fixture to determine conformance with the requirements of 3.2 through 3.4.2 and 3.7 through 3.9. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied to the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the fixture and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.3), shall be cause for rejection.

4.6 Tests. Each fixture shall be tested to determine compliance with test requirements as specified in 4.6.1 through 4.6.6.

4.6.1 Key seat cutting. The fixture shall be mounted on a lathe. Using a No. 204 Woodruff cutter (or equal) and a No. 608 Woodruff cutter (or equal), respectively, one key seat shall be cut in a three-quarter inch diameter bar and in a two inch diameter bar of AISI 1045 steel. Key seats shall be examined for compliance with 3.5.1.1.

4.6.2 Keyway cutting. The fixture shall be mounted on a lathe. Using a minimum feed of 1 3/4 inches per minute, one keyway measuring 1/4 inch deep x 1/2 inch wide x 6 inches long shall be cut in a two inch round bar of AISI 1045 steel. The keyway shall be examined/tested for compliance with 3.5.1.2.

4.6.3 Vertical slotting. The fixture shall be mounted on a lathe. One vertical slot measuring 0.125 inches deep, 0.375 inches wide, and one inch long shall be cut in a 1/4 inch deep by 1 1/2 inch wide by 6 inches long piece of AISI 1045 steel. Accuracy shall be as specified in 3.5.1.3.

4.6.4 Rigidity. The basic fixture shall be mounted on a machine of sufficient size, rigidity, and horsepower to conduct this test. A test specimen, a one inch square bar of AISI 1045 cold rolled steel, shall be mounted on the fixture. Using a one inch diameter end mill, a cut shall be taken, as specified in 3.5.3. The test specimen shall then be examined for compliance with 3.5.3.

4.6.5 Indexing. The indexing device and basic fixture shall be mounted on a lathe, and test cuts shall be made on the periphery of a two inch diameter bar made of AISI 1045 cold rolled steel. Cuts shall be made one-eighth inch wide, one-eighth

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inch deep, and one inch long. Cuts shall be made at 45 degrees, 90 degrees, 135 degrees, 180 degrees, and 270 degrees from the first cut l The cuts shall be examined for compliance with 3.5.2.

4.6.6 Protective finish. The hardness, adhesion, and thickness of the paint used on the fixture shall be checked as follows:

a. The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the fixture until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

b. A test sample of the same metal, painted with the same process and at the same time as the fixture, shall have a sharpened H lead pencil pressed into the paint as in a. Failure of the surface to pit (paint too hard) shall be cause for rejection.

c. Two, 2 inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection. (see 3.6).

d. Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 inch thick shall be cause for rejection (see 3.6).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with Section 5 requirements.

5. PACKAGING

5.1 Preservation.Preservation shall be Level A in accordance with PPP-P-40 or Level C in accordance with MIL-STD-1190 as specified (see 6.2.1).

5.2 Packing. Packing shall be Level A or B in accordance with PPP-P-40 or Level C in accordance with MIL-STD-1190 as specified (see 6.2.1).

5.3 Marking. Marking shall be in accordance with MIL-STD-129, and shall include bar code marking in accordance with MIL-STD-1189.

6. NOTES

6.1 Intended use. The lathe fixture covered by this specification is intended for use in milling, drilling, slotting, tapping, layout, boring, turning, and indexing operations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. First article inspection when required (see 3.1).

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c. Standard equipment and accessories if different (see 3.7).

d. Identification marking if different (see 3.8).

e. First article inspection if different (see 4.3).

f. Quality conformance inspection if different (see 4.4).

g. Sampling when required (see 4.4.1).

h. Selection of applicable level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article (see 4.3).

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and controls of noise levels, radiation, electromagnetic emissions, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.6 Repair part manuals. Repair parts and accessory identification in repair part manuals shall include the prime contractor's name, nomenclature and part number. Repair part identification in repair part manuals shall also include the original equipment/part manufacturer's (OEM) (actual part manufacturer's) name, Commercial and Government Entity (CAGE), nomenclature and part number of the part used in the end item, immediately after the prime contractor's part identification. Items with high mortality rates shall be so identified, see paragraph 3.1.2.2 thru 3.1.2.10 (especially paragraph 3. 1.2.8.2.2) of MIL-M-7298C.

6.7 Subject term (key word)listing.

Keyway cutting
Vertical slotting
Indexing device

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AL

Review activities:
Army - SM

Preparing activity:
Army - AL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-F-45569C(AL) Fixture, Lathe, Metal CUtting

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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